

Smart Countdown – Task 5 (7-Segment Display + Buzzer)

Overview

This Arduino sketch displays a countdown from **9 to 0** on a 7-segment display while sounding a short beep at each count, finishing with a longer completion tone.

Hardware

Component	Arduino Pin
Buzzer	8
Segment a	2
Segment b	3
Segment c	4
Segment d	5
Segment e	6
Segment f	7

Component	Arduino Pin
Segment g	9

Pin 8 is reserved for the buzzer, so segment pins skip from 7 to 9.

How It Works

1. Setup

- Initializes Serial communication at 9600 baud.
- Configures the buzzer pin and all 7 segment pins as `OUTPUT` .

2. Countdown Loop (in `setup()`)

- Starts at `count = 9` and loops while `count > 0` :
 - Prints the current count to Serial.
 - Calls `showDigit(count)` to update the display.
 - Plays a short 200 ms beep at 1000 Hz.
 - Waits 1 second (`delay(1000)`).
 - Decrements `count` .

3. Completion

- Displays `0` on the segment display.

- Plays a longer 1000 ms tone at 1500 Hz to signal the countdown is complete.
- Prints `"=== Countdown Complete ==="` to Serial.

4. `loop()`

- Left empty — the countdown runs once per power-up/reset.

Key Functions

`showDigit(int n)`

Looks up the segment pattern for digit `n` in the `digits[10][7]` array and writes `HIGH / LOW` to each segment pin accordingly. Includes a safety check to ignore invalid input (`n < 0` or `n > 9`).

`digits[10][7]`

A lookup table where each row represents which of the 7 segments (a–g) should be lit to display digits 0–9.

Notes / Possible Improvements

- Remove the stray double semicolon after the completion `tone()` call (`;;` → `;`).

- Since `loop()` is empty, the countdown only runs once. To re-trigger it (e.g., via a button press), move the countdown logic into `loop()` guarded by a trigger condition.
- Consider using `millis()` instead of `delay()` if you want the sketch to remain responsive to other inputs during the countdown.

Serial Monitor Output Example

```
=== Beeping Countdown Starting===  
Counting: 9  
Counting: 8  
Counting: 7  
Counting: 6  
Counting: 5  
Counting: 4  
Counting: 3  
Counting: 2  
Counting: 1  
=== Countdown Complete ===
```